C3 Quantitative Chemistry – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 1: Mass and Moles **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Law of conservation of mass				Mass of products = mass of reactants
2	Relative formula mass (M_r)				Sum of the relative atomic masses in the formula
3	Why mass increases when a metal is heated in air				Oxygen from the air is added
4	Why mass decreases when a carbonate is heated				CO_2 gas escapes
5	Mole				Amount containing 6.02×10^{23} particles
6	Avogadro constant				6.02×10^{23} per mole
7	Equation for moles				Moles = mass ÷ M_r
8	Mass of 1 mole of a substance				Its M_r in grams
9	Limiting reactant				Reactant completely used up
10	Reactant in excess				Reactant left over

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C3 Quantitative Chemistry – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 2: Concentration and Measurement **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Concentration equation				Mass of solute ÷ volume of solution
2	Units of concentration				g/dm^3
3	1 dm^3 in cm^3				1000 cm^3
4	Convert cm^3 to dm^3				÷ 1000
5	Mass of solute from concentration				Concentration × volume
6	Numbers in a balanced equation show				Mole ratio
7	Mean				Total ÷ number of values
8	Range				Highest – lowest value
9	Uncertainty				Range of results about the mean
10	Effect of more solute in the same volume				Higher concentration

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C3 Quantitative Chemistry – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 3: Yield, Atom Economy, Titrations and Gases **Paper 1** **Triple only**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Percentage yield equation				Actual mass ÷ maximum theoretical mass × 100
2	Reasons yield is below 100%				Reversible reaction, losses when separating, side reactions
3	Atom economy equation				M_r of desired product ÷ total M_r of reactants × 100
4	Why high atom economy matters				Less waste; more sustainable
5	Concentration in mol/dm^3				Moles ÷ volume in dm^3
6	Glassware to measure 25.0 cm^3 of alkali				Pipette
7	Glassware to add acid in a titration				Burette
8	Volume of 1 mole of any gas at RTP				24 dm^3
9	Room temperature and pressure (RTP)				20 °C and 1 atm
10	Equation for gas volume at RTP				Moles × 24 dm^3

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.